

FEBAUTO Racing

Member guide, release 2

Part I The box: your ROS 2 environment

Part II Drivers: write, score, submit

Part III The car

Formula Electric at Berkeley

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https://github.com/FEBAutonomous/feb-racing

https://febautonomous.github.io/feb-racing/

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1 What changed, in one minute

Release 2 replaces the throwaway devkit container with **the box**: a Linux machine with ROS 2 Humble inside Docker that keeps what you install, sees your whole FEB folder, and runs the same on the car's Jetson. Your driver talks the real car's topics, so what laps in the simulator drives the car unchanged. The launcher is `feb`; `feb-sim` is gone.

Was	Is
`./feb-sim run`	`./feb up`
`./feb-sim shell`	`./feb shell`, or `./feb code` for VS Code inside the box
container wiped on every run	the box persists; `./feb reset` wipes it on purpose
only `stack/` visible inside	all of `~/FEB` visible inside
topics `/autodrive/roboracer_1/*`, throttle in $[0, 1]$	`/scan`, `/odom`, `/imu`, camera in; `/drive` (m/s, rad) out
one practice track	every practice track of your driver's kind
submit posts a result	submit posts the results and your code

Part I
The box

2 Setup

Install these three, once:

1. **Docker.** Mac: Docker Desktop or OrbStack. Windows: Docker Desktop, with *WSL integration* turned on for your Ubuntu in its settings. Linux: `sudo apt install docker.io`, add yourself to the `docker` group, log out and in.
2. **VS Code** with the *Dev Containers* extension. Cursor works the same way.
3. **Foxglove** from `https://foxglove.dev/download`. Optional; the browser desktop covers most of it.

Windows: everything below happens in a **WSL 2 Ubuntu terminal**, never PowerShell, and `~/FEB` stays inside WSL. No Ubuntu yet: `wsl --install -d Ubuntu` in PowerShell, reboot, open "Ubuntu" from the Start menu. Mac: keep `~/FEB` under `/Users`.

Had the old devkit? Release 2 rewrote the repository, so the old clone cannot be updated; delete it. If you wrote code in its `stack/` folder, copy that folder somewhere first and drop it into the new `stack/` after setup; it works again once it has a card (section 5).

```
rm -rf ~/feb-racing            # wherever the old clone is; skip if you never had it
```

Then the setup itself:

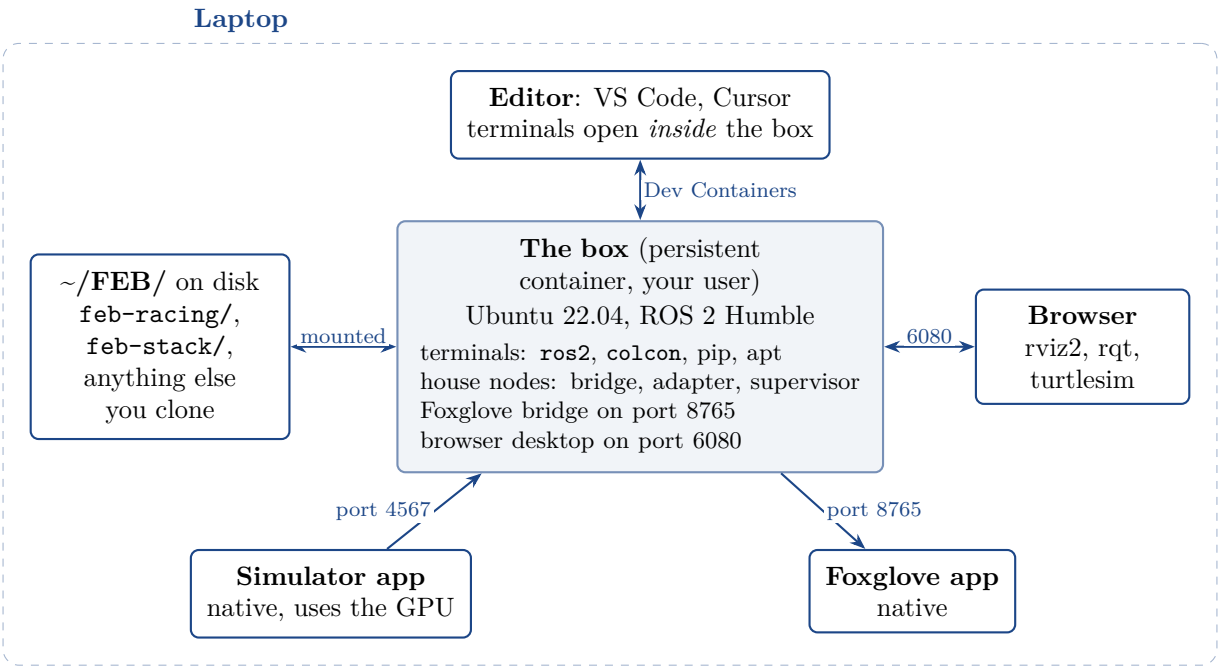
```
mkdir -p ~/FEB && cd ~/FEB
git clone git@github.com:FEBAutonomous/feb-racing.git && cd feb-racing
./feb setup --name "Your Name"
```

`setup` pulls the image (a few GB, once), downloads the simulator app for your OS, and creates your box. Your name is how results appear on the scoreboard; ask the lead to add it to `submissions.yaml` with your GitHub login. No SSH key on GitHub yet: use `https://github.com/FEBAutonomou` in the clone line.

First drive: `./feb up`. The simulator opens, arrow keys drive, close the window to stop. `./feb up --driver feb_driver` races the starter driver instead. Mac, first time only: if macOS refuses to open the simulator, right-click the app in `~/Library/Application Support/feb-sim/app` and choose Open.

This is a one-time clone. Every later release arrives with `./feb update`.

3 How it fits together



The box is one container, made once by `setup`, started by whichever command needs it, kept until `./feb reset`. Your `~/FEB` folder is mounted into it at `/home/feb/FEB`: the same files, not a copy, so edit on the laptop with anything or inside the box, it makes no difference. What you `apt` or `pip` install lives in the box and goes with a reset or a new image; your files never do.

4 The commands

Command	What it does
`./feb shell`	a terminal in the box, in the folder you ran it from; open as many as you like
`./feb shell --gui`	the same, plus the box's desktop in a browser tab (rviz2, rqt, turtlesim)
`./feb code`	VS Code attached to the box (`--cursor` for Cursor)
`./feb up`	house nodes, simulator, Foxglove; drive by hand; Ctrl-C ends it
`./feb up --driver NAME`	your driver races; `--track T`, `--mode-name M`, `--noise 1`, `--opponent normal`
`./feb up --deadman --driver NAME`	the car's rules in the simulator (section 7)
`./feb stop`	ends `feb up` from another terminal; the box stays
`./feb logs -f`	what `feb up` printed
`./feb practice --driver NAME`	scored runs on every practice track of the driver's kind
`./feb submit`	results to the scoreboard, code to the team repo, one pull request each
`./feb update`	git pull, newer image (replaces the box), newer simulator
`./feb reset`	a fresh box; your files are untouched

Inside the box there is nothing of ours to learn: `ros2`, `colcon`, `pip`, `ros2 bag record -a`, and `rviz2 &` which lands on the browser desktop.

rviz in the browser. With `./feb up` running in one terminal, in a second one: `./feb shell --gui`, then `rviz2 &`. A browser tab opens on the box's desktop and rviz appears there. Fixed Frame `odom`; Add, By topic, `/scan` (LaserScan, style Points, size 4) and `/odom`. A topic that stays empty wants Reliability Policy set to Best Effort. **Foxglove app:** Open connection, Foxglove WebSocket, `ws://localhost:8765`; the bridge runs whenever `feb up` does.

Part II
Drivers

5 A driver is one package

Copy the template and rename it:

```
cp -r starter_kit/feb_driver stack/my_driver
```

File	Holds
`my_driver/reactive_driver.py`	the node: scan in, speed and steering out
`config/driver.yaml`	every tuning number, one per line with a comment
`config/driver.car.yaml`	only the numbers that differ on the real car; loaded on top, on the car only
`launch/drive.launch.py`	starts the node with the yaml; includes `feb_mission` if present
`driver.yaml`	the driver's card (below)

```
name: my_driver
kind: walls                 # walls | cones | any
maintainer: your name
packages: [my_driver]       # what submit copies: your own packages
modes:
  trackdrive: {launch: drive.launch.py, max_speed: 3.0}
  inspection: {launch: drive.launch.py, max_speed: 0.8}
default_mode: trackdrive
```

A mode is a launch file plus a speed cap; `inspection` is the slow first lap of anything new, `trackdrive` the race. Edit the Python, save, run again: no rebuild for Python. Keep the package in git, `stack/` is ignored by this repo on purpose.

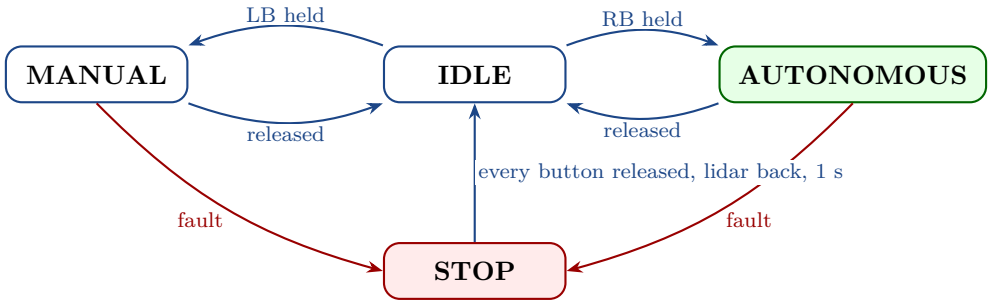
6 The contract

Topic	Type	
`/scan`	LaserScan	the lidar, 1080 beams over 270°, frame `laser`
`/odom`	Odometry	pose and speed, dead-reckoned from the wheels and the gyro
`/imu`	Imu	
`/camera/color/image_raw`	Image	the front camera (cone tracks)
`/drive`	AckermannDriveStamped	**yours**: `drive.speed` in m/s, `drive.steering_angle` in rad, at most ±0.5236
`/feb/state`	String	what the supervisor is doing: IDLE, MANUAL, AUTONOMOUS, STOP

These are the RoboRacer car's topics. In the simulator an adapter provides them; on the car the VESC and lidar drivers do. A command older than half a second stops the car. The simulator's own `/autodrive/...` topics remain for debugging; the car does not have them.

7 The dead-man, and the supervisor

On the car nothing moves unless a button on the gamepad is held. The supervisor is the node that enforces it; it sits between every driver and the wheels, in the simulator and on the car alike.



fault: lidar silent for 0.5 s, the X button, or a collision imminent

- **IDLE**: the car sits. Your stack is running and publishing `/drive`; it goes nowhere.
- **MANUAL**, LB held: the sticks drive. For positioning the car and checking wiring.
- **AUTONOMOUS**, RB held: your `/drive` reaches the wheels, clipped to the mode's cap.
- **STOP**: brakes. Entered when the lidar goes silent, the stop button is pressed, or a collision is imminent; left for IDLE only once every button is released and the sensors are back.
- Release either button: zero speed, which is a brake, within a third of a second.

In the simulator, `./feb up --deadman --driver NAME` runs the same rules with the keyboard as the pad: hold SPACE for RB, hold M for LB with the arrows driving, X for the stop button, Q to quit. The nodes' output goes to `./feb logs -f` in another terminal. Run every driver this way once before it goes near the car.

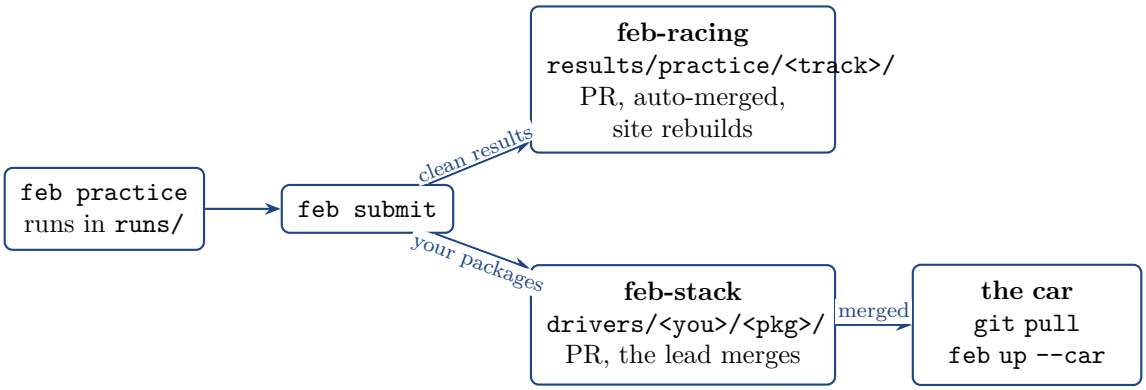
Your stack and the state. The button never starts or stops your stack; it only gates its output. So anything that remembers, an MPC's warm start, a localiser, an integrator, is stale when RB goes down. The rule: *on AUTONOMOUS, start clean.* `feb_mission`, a forty-line package in the starter kit, turns `/feb/state` into two hooks, `on_go` and `on_stop`, and a live flag; a stack with memory fills them in. The gap follower needs none of it.

8 Score and submit

```
./feb practice --driver my_driver        # every practice track of its kind, competition
    rules
./feb submit --dry-run                   # what would be posted
./feb submit
```

`practice` runs each track in its own clean container the way the competition does: a warm-up lap and ten timed laps, 300 s limit, 10 s per collision, headless at the 10 Hz scoring rate. Each run lands in `runs/<id>/` with a result file, a bag and the logs.

`submit` does two things from one command. It posts every clean run (finished, no collisions) not yet posted to the scoreboard as one pull request to `feb-racing`, which checks and merges itself; the site rebuilds and your time appears, unverified until the nightly rerun. And it copies your driver's packages, the `packages:` list in its card, into the team repo `feb-stack` under `drivers/<your name>/`, on a branch named after you, with a pull request to main that the lead reviews. Nothing changed since last time: nothing is pushed. `--no-code` and `--no-results` pick one half.



The team repo. `FEBAutonomous/feb-stack` holds the drivers that drive, one folder per person under `drivers/`, and one copy of the shared packages under `shared/` (`feb_mission`, `feb_cone_ordering`). Nobody edits it by hand. To run a teammate's merged driver in your simulator: `./feb up --ws ~/FEB/feb-stack --driver theirs`.

Part III
The car

9 From a merged driver to the wheels

The Jetson on the car has the same box, the same `~/FEB` folder, and the pad's dongle. Whoever runs the car, over ssh:

```
cd ~/FEB/feb-stack && git pull
cd ~/FEB/feb-racing
./feb up --car --driver my_driver --mode-name inspection
```

On the car the workspace is `feb-stack` by default, so every merged driver is built and the named one launches: the VESC, lidar and gamepad drivers instead of the simulator, the supervisor with the dead-man required and the mode's cap, then your launch file with your car yaml on top. LB held, a human drives; RB held, your code drives; let go, it stops. Foxglove on a laptop at `ws://<the car>:8765` shows the same panels as in the simulator. First drive in `inspection`, then `trackdrive`. The lead's own notes on the car, wiring to first drive, are `docs/build-car.md`.

10 When something is wrong

Symptom	Fix
`Cannot connect to the Docker daemon`	Docker Desktop is not running; open it, wait for the whale icon to settle
`this clone sits directly in /home/...`	the clone needs a folder of its own: `mkdir ~/FEB && mv ~/feb-racing ~/FEB/`
`unauthorized` while pulling the image	a stale `ghcr.io` login: `docker logout ghcr.io`, run setup again
`refusing to merge unrelated histories`	you pulled in the old clone; delete it and clone again (Setup)
`no driver.yaml with name '...'`	the package has no card, or its `name:` differs: section 5
`ros2: command not found`	you are in a laptop terminal: `./feb shell`
my node is not there	it did not build: `./feb logs -n 100`, or `colcon build` in the workspace and read it
the simulator says not connected	the house nodes are still starting; `./feb logs -f`
`port is already allocated`	another box holds the port: `./feb reset`
`mounts denied` (Mac)	`~/FEB` is outside `/Users`: move it
the browser desktop will not connect	reload; still nothing, `./feb reset`
nothing is clean enough to submit	a run with a collision is not posted by default; drive slower, or `--all`
anything weird	`./feb reset`: thirty seconds, nothing of yours is in there